EE4-66 Topics in Large Dimensional Data Processing

Instructions for Candidates

Answer all three questions. Each question carries 25 marks.

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1. Let $\boldsymbol{I}_4 \in \mathbb{R}^{4\times 4}$ be the 4×4 identity matrix and

$$\boldsymbol{H} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}.$$

It is clear that $\boldsymbol{H}^\mathsf{T}\boldsymbol{H} = \boldsymbol{H}\boldsymbol{H}^\mathsf{T} = \boldsymbol{I}_4$. Denote the i-th column of $\boldsymbol{H}$ by $\boldsymbol{H}_{:,i}$.

(a) Let

$$\boldsymbol{A} = [\boldsymbol{H}_{:,1}, \boldsymbol{H}_{:,2}]\begin{bmatrix} 1/4 & 0 \\ 0 & 1 \end{bmatrix}.$$

i Compute $\boldsymbol{A}^\mathsf{T}\boldsymbol{A}$. Find the singular value decomposition (SVD) of $\boldsymbol{A}$. Based on the SVD, compute the Moore-Penrose pseudo-inverse of $\boldsymbol{A}$, Denoted by $\boldsymbol{A}^\dagger$. [3]

ii Consider a linear system

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{w}$$

where $\boldsymbol{w}$ represents random noise. For a given measurement vector $\boldsymbol{y}$, the optimal estimate of the signal (in the least squares sense) is given by

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2.$$

Let

$$\boldsymbol{y}_1 = [-0.275,\ -0.275,\ 0.725,\ 0.725]^\mathsf{T},$$
$$\boldsymbol{y}_2 = [-0.275,\ -0.475,\ 0.525,\ 0.725]^\mathsf{T}.$$

Compute the corresponding least squares solutions $\boldsymbol{x}_1^\star$ and $\boldsymbol{x}_2^\star$ respectively.

Both measurement vectors $\boldsymbol{y}_1$ and $\boldsymbol{y}_2$ are actually generated from the same signal $\boldsymbol{x}_0$. More precisely, $\boldsymbol{y}_1 = \boldsymbol{A}\boldsymbol{x}_0 + \boldsymbol{w}_1$ and $\boldsymbol{y}_2 = \boldsymbol{A}\boldsymbol{x}_0 + \boldsymbol{w}_2$ with $\|\boldsymbol{w}_1\|_2 = \|\boldsymbol{w}_2\|_2$, $\boldsymbol{w}_1 \in \mathit{span}(\boldsymbol{A})$, *and* $\boldsymbol{w}_2 \perp \mathit{span}(\boldsymbol{A})$. Find $\boldsymbol{x}_0$, $\boldsymbol{w}_1$, and $\boldsymbol{w}_2$. Comment on how different noise vectors with the same

ℓ_2-norm can affect the estimation result differently. [4]

Solution: It is clear that

$$\boldsymbol{A}^\mathsf{T}\boldsymbol{A} = \begin{bmatrix} 1/16 & 0 \\ 0 & 1 \end{bmatrix}.$$

SVD of $\boldsymbol{A}$ is given by

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{H}_{:,1} & \boldsymbol{H}_{:,2} \end{bmatrix} \begin{bmatrix} 1/4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Moore-Penrose pseudo-inverse of $\boldsymbol{A}$ is therefore

$$\begin{aligned} \boldsymbol{A}^\dagger &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{H}_{:,1} & \boldsymbol{H}_{:,2} \end{bmatrix}^\mathsf{T} \\ &= \begin{bmatrix} 2 & 2 & 2 & 2 \\ 1/2 & 1/2 & -1/2 & -1/2 \end{bmatrix} \end{aligned}$$

The least squares solutions corresponding to $\boldsymbol{y}_1$ and $\boldsymbol{y}_2$ are given by

$$\begin{aligned} \boldsymbol{x}_1^\star &= \boldsymbol{A}^\dagger \boldsymbol{y}_1 = [1.8,\ -1]^\mathsf{T}, \\ \boldsymbol{x}_2^\star &= \boldsymbol{A}^\dagger \boldsymbol{y}_2 = [1,\ -1]^\mathsf{T}. \end{aligned}$$

Based on the assumption $\boldsymbol{w}_2 \perp \text{span}(\boldsymbol{A})$, it can be verified that

$$\begin{aligned} \boldsymbol{x}_0 &= [1,\ -1]^\mathsf{T}, \\ \boldsymbol{w}_2 &= [0.1,\ -0.1,\ -0.1,\ 0.1]^\mathsf{T}. \end{aligned}$$

Note that $\boldsymbol{w}_1 \in \text{span}(\boldsymbol{A})$. It holds that

$$\boldsymbol{w}_1 = [0.1,\ 0.1,\ 0.1,\ 0.1]^\mathsf{T}.$$

From this example, a noise vector along the singular vector corresponding to a non-zero singular value s will be 'amplified' by $1/s$. A noise vector along the singular vector corresponding to a zero singular value s will be nullified.

(b) Let

$$\boldsymbol{A} = \Big[\boldsymbol{H},\ \boldsymbol{I}_4\Big] \in \mathbb{R}^{4\times 8}.$$

i Compute $\boldsymbol{A}\boldsymbol{A}^\mathsf{T}$, the SVD of $\boldsymbol{A}$, and the Moore-Penrose pseudo-inverse of $\boldsymbol{A}$. [2]

ii Compute the mutual coherence constant μ of $\boldsymbol{A}$. [2]

iii Compute the restricted isometry constant δ_2 of $\boldsymbol{A}$. [3]

iv Consider a noise-free linear system

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}.$$

Suppose that

$$\boldsymbol{y} = [2,\ 1,\ 1,\ 2]^\mathsf{T}.$$

A. Find the least squares solution

$$\boldsymbol{x}_{\text{ls}} = \arg\min_{\boldsymbol{x}}\ \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2.$$

[2]

B. Find the sparsest solution

$$\boldsymbol{x}_{\text{sp}} = \arg\min_{\boldsymbol{x}}\ \|\boldsymbol{x}\|_0 \text{ s.t. } \boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}.$$

[2]

C. If

$$\boldsymbol{y} = [2,\ 0,\ 0,\ 2]^\mathsf{T},$$

Find the sparsest solution. Is the sparsest solution unique? [2]

Solution:

$$\boldsymbol{A}\boldsymbol{A}^\mathsf{T} = \boldsymbol{H}\boldsymbol{H}^\mathsf{T} + \boldsymbol{I}_4 * \boldsymbol{I}_4^\mathsf{T} = 2\boldsymbol{I}_4.$$

SVD of $\boldsymbol{A}$ is given by

$$\boldsymbol{A} = \underbrace{\boldsymbol{I}_4}_{\boldsymbol{U}} \ \underbrace{\sqrt{2}\boldsymbol{I}_4}_{\boldsymbol{S}} \ \underbrace{\frac{1}{\sqrt{2}}\boldsymbol{A}}_{\boldsymbol{V}^\mathsf{T}}.$$

The Moore-Penrose pseudo-inverse of $\boldsymbol{A}$ is

$$\begin{aligned}\boldsymbol{A}^\dagger &= \frac{1}{\sqrt{2}}\boldsymbol{A}^\mathsf{T} \ \frac{1}{\sqrt{2}}\boldsymbol{I}_4 \ \boldsymbol{I}_4^\mathsf{T} \\ &= \frac{1}{2}\boldsymbol{A}^\mathsf{T}.\end{aligned}$$

The mutual coherence constant μ is given by

$$\mu = \max_{i \neq j} |\langle \boldsymbol{a}_i, \boldsymbol{a}_j \rangle| = 0.5.$$

By the definition of RIC δ_2 and the structure of $\boldsymbol{A}$, it can be verified that

$$\begin{aligned}\delta_2 &= \max_i \left| s_i \left([\boldsymbol{a}_1, \ \boldsymbol{a}_5]^\mathsf{T} [\boldsymbol{a}_1, \ \boldsymbol{a}_5] \right) - 1 \right| \\ &= \max_i \left| s_i \left(\begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} \right) - 1 \right| \\ &= \max(|1.5 - 1|, |0.5 - 1|) = 0.5.\end{aligned}$$

For

$$\boldsymbol{y} = [2, \ 1, \ 1, \ 2]^\mathsf{T},$$

the least squares solution is given by

$$\begin{aligned}\boldsymbol{x}_{\text{ls}} &= \boldsymbol{A}^\dagger \boldsymbol{y} \\ &= [1.5, \ 0, \ 0.5, \ 0, \ 1, \ 0.5, \ 0.5, \ 1]^\mathsf{T}.\end{aligned}$$

and the sparsest solution is given by

$$\boldsymbol{x}_{\text{sp}} = [3, \ 0, \ 1, \ 0, \ 0, \ 0, \ 0, \ 0]^\mathsf{T}.$$

For

$$\boldsymbol{y} = [2,\ 0,\ 0,\ 2]^{\mathsf{T}},$$

the sparsest solution is not unique:

$$\boldsymbol{x}_{\text{sp},1} = [2,\ 0,\ 2,\ 0,\ 0,\ 0,\ 0,\ 0]^{\mathsf{T}},\ \text{and}$$
$$\boldsymbol{x}_{\text{sp},2} = [0,\ 0,\ 0,\ 0,\ 2,\ 0,\ 0,\ 2]^{\mathsf{T}}$$

can generate $\boldsymbol{y}$.

(c) (Low rank matrix recovery problems)

i Let $\boldsymbol{X}$ be a rank-one matrix. Its partial observations are given by

$$\boldsymbol{X}_\Omega = \begin{bmatrix} 1 & ? \\ -1 & ? \\ -1 & 1 \\ ? & -1 \end{bmatrix}.$$

Find $\boldsymbol{X}$. [2]

ii Let $\boldsymbol{X}$ be a rank-two matrix. Its partial observations are given by

$$\boldsymbol{X}_\Omega = \begin{bmatrix} 3 & 5 & 3 & ? \\ 5 & 3 & ? & 3 \\ 3 & ? & 3 & 5 \\ ? & 3 & 5 & 3 \end{bmatrix}.$$

Find $\boldsymbol{X}$. [3]

Solution:

i As $\boldsymbol{X}$ is a rank-one matrix, it can be written as $\boldsymbol{X} = \boldsymbol{u}\boldsymbol{v}^{\mathsf{T}}$ where both $\boldsymbol{u}$ and $\boldsymbol{v}$ are vectors, meaning that both columns of $\boldsymbol{X}$ are scaled versions of the same vector $\boldsymbol{u}$. From the observed entries, it

is clear that $\boldsymbol{u} = [1,\ -1,\ -1,\ 1]^\mathsf{T}$. Hence $\boldsymbol{X}$ is given by

$$\boldsymbol{X} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ -1 & 1 \\ 1 & -1 \end{bmatrix}.$$

ii The submatrix $\boldsymbol{X}_{1:2,1:2}$ already has rank two and can be written as

$$\begin{aligned} \boldsymbol{X}_{1:2,1:2} &= \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} + \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \\ &= 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}^\mathsf{T} - \begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}^\mathsf{T}. \end{aligned}$$

From $\boldsymbol{X}_{3:4,3:4}$ and other observed entries, it can be verified that

$$\begin{aligned} \boldsymbol{X} &= 4 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}^\mathsf{T} - \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}^\mathsf{T} \\ &= \begin{bmatrix} 3 & 5 & 3 & 5 \\ 5 & 3 & 5 & 3 \\ 3 & 5 & 3 & 5 \\ 5 & 3 & 5 & 3 \end{bmatrix}. \end{aligned}$$

2. (a) i Let $f(x)$ be any convex function. Let $g(x) = ax + b$ be any affine function. Is $h(x) = f(g(x))$ convex or not? Prove your answer. [3]

ii Let $f(x) = x^2$, $g(x) = |x| - 3$, and $h(x) = f(g(x))$. Sketch the function $h(x)$. Based on your sketch, is $h(x) = f(g(x))$ is convex or not. [2]

Solution:

i $h(x)$ is convex because

$$\begin{aligned} h(\lambda x_1 + (1-\lambda)x_2) &= f(g(\lambda x_1 + (1-\lambda)x_2)) \\ &= f(\lambda g(x_1) + (1-\lambda)g(x_2)) \\ &\leq \lambda f(g(x_1)) + (1-\lambda)f(g(x_2)) \\ &= \lambda h(x_1) + (1-\lambda)h(x_2), \end{aligned}$$

where the 2nd equality follows from the fact that $g(x)$ is linear in x.

ii It is easy to see from the sketch that $h(x)$ is not convex.

(b) i Let

$$f(\boldsymbol{x}) = \frac{1}{2}\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2 + \frac{1}{2}\|\boldsymbol{C}\boldsymbol{x} - \boldsymbol{d}\|_2^2,$$

where

$$\boldsymbol{A} = \begin{bmatrix} 2 & -3 \\ 0 & 3 \end{bmatrix},\ \boldsymbol{b} = \begin{bmatrix} -1 \\ 2 \end{bmatrix},\ \boldsymbol{C} = \begin{bmatrix} 2 & 0 \\ 1 & 6 \end{bmatrix},\ \boldsymbol{d} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}.$$

Compute the gradient of $f(\boldsymbol{x})$, denoted by $\nabla f(\boldsymbol{x})$. Find the optimal solution $\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}}\ f(\boldsymbol{x})$. [3]

ii Let

$$f(\boldsymbol{x}) = \|\boldsymbol{x} - \boldsymbol{b}\|_1 + \frac{1}{6}\|\boldsymbol{x} - \boldsymbol{d}\|_2^2,$$

where

$$\boldsymbol{b} = \begin{bmatrix} 0 \\ 6 \end{bmatrix},\ \boldsymbol{d} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

Compute the sub-differential of $f(\boldsymbol{x})$, denoted by $\partial f(\boldsymbol{x})$. Find the op-

timal solution $\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} \ f(\boldsymbol{x})$. [5]

Solution:

i

$$\begin{aligned}\nabla f(x) &= \boldsymbol{A}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} - \boldsymbol{A}^\mathsf{T}\boldsymbol{b} + \boldsymbol{C}^\mathsf{T}\boldsymbol{C}\boldsymbol{x} - \boldsymbol{C}^\mathsf{T}\boldsymbol{d} \\ &= (\boldsymbol{A}^\mathsf{T}\boldsymbol{A} + \boldsymbol{C}^\mathsf{T}\boldsymbol{C})\boldsymbol{x} - (\boldsymbol{A}^\mathsf{T}\boldsymbol{b} + \boldsymbol{C}^\mathsf{T}\boldsymbol{d}) \\ &= \left(\begin{bmatrix} 4 & -6 \\ -6 & 18 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 6 & 36 \end{bmatrix}\right)\boldsymbol{x} - \left(\begin{bmatrix} -2 \\ 9 \end{bmatrix} + \begin{bmatrix} 1 \\ -18 \end{bmatrix}\right) \\ &= \begin{bmatrix} 9 & 0 \\ 0 & 54 \end{bmatrix}\boldsymbol{x} + \begin{bmatrix} 1 \\ 9 \end{bmatrix}.\end{aligned}$$

Set $\nabla f(\boldsymbol{x}) = \boldsymbol{0}$. One has

$$\boldsymbol{x}^\star = -\begin{bmatrix} 1/9 \\ 1/6 \end{bmatrix} = -\begin{bmatrix} 0.111 \\ 0.167 \end{bmatrix}.$$

ii

$$\partial f(\boldsymbol{x}) = \begin{bmatrix} \operatorname{sign}(x_1) \\ \operatorname{sign}(x_2 - 6) \end{bmatrix} + \frac{1}{3}\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 5 \\ 1 \end{bmatrix}\right),$$

where

$$\operatorname{sign}(x) = \begin{cases} -1 & \text{if } x < 0, \\ [-1, 1] & \text{if } x = 0, \\ 1 & \text{if } x > 0. \end{cases}$$

The optimal $\boldsymbol{x}^\star$ should satisfy $\boldsymbol{0} \in \partial f(\boldsymbol{x}^\star)$. Hence

$$\boldsymbol{x}^\star = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

(c) In the k-th iteration of gradient descent method, the operation

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \tau\nabla f(\boldsymbol{x}^k)$$

is performed where $\tau > 0$ is an appropriately chosen step size and $\nabla f(\boldsymbol{x}^k)$

is the gradient of f at the point $\boldsymbol{x}^k$.

i Let

$$g(\boldsymbol{x}) = \frac{1}{2}\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2.$$

Find the first order Taylor approximation of $g(\boldsymbol{x})$ at the point $\boldsymbol{x}^k$. Based on it, find

$$\lim_{\tau \to 0} \frac{g(\boldsymbol{x}^k - \tau \nabla g(\boldsymbol{x}^k)) - g(\boldsymbol{x}^k)}{\tau}.$$

Using your result, state whether $g(\boldsymbol{x}^k - \tau \nabla g(\boldsymbol{x}^k)) \leq g(\boldsymbol{x}^k)$ or not when the step size τ is sufficiently small. [3]

ii Let $\boldsymbol{x} \in \mathbb{R}^2$, and x_1 and x_2 denote the first and second entries of $\boldsymbol{x}$ respectively. Let

$$h(\boldsymbol{x}) = |x_1| + 4|x_2|.$$

Consider the point

$$\boldsymbol{x}^k = [1, 0]^\mathsf{T}.$$

Find the sub-differential of h at $\boldsymbol{x}^k$, denoted by ∂h. Among all sub-gradients $\boldsymbol{g} \in \partial h$, choose the one of which both elements attain the maximum values. Denote it by $\boldsymbol{g}^{\max}$. Using the definition of h (not Taylor approximation this time as it does not exist), compute

$$\lim_{\tau \to 0} \frac{h(\boldsymbol{x}^k - \tau \boldsymbol{g}^{\max}) - h(\boldsymbol{x}^k)}{\tau}.$$

Using your result, state whether $h(\boldsymbol{x}^k - \tau \boldsymbol{g}^{\max}) \leq h(\boldsymbol{x}^k)$ or not when the step size τ is sufficiently small.
What does this example imply for sub-gradient descent method in general? [5]

Solution:

i In the neighbourhood of $\boldsymbol{x}^k$, the first order Taylor approximation is

given by

$$\begin{aligned} g(\boldsymbol{x}) &\approx g(\boldsymbol{x}^k) + \left\langle \nabla g(\boldsymbol{x}^k), \boldsymbol{x} - \boldsymbol{x}^k \right\rangle \\ &= g(\boldsymbol{x}^k) + \left\langle \boldsymbol{A}^\mathsf{T}(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{b}), \boldsymbol{x} - \boldsymbol{x}^k \right\rangle. \end{aligned}$$

Hence

$$\begin{aligned} &\lim_{\tau \to 0} \frac{g(\boldsymbol{x}^k - \tau \nabla g(\boldsymbol{x}^k)) - g(\boldsymbol{x}^k)}{\tau}. \\ &= \lim_{\tau \to 0} \frac{\left\langle \nabla g(\boldsymbol{x}^k), -\tau \nabla g(\boldsymbol{x}^k) \right\rangle}{\tau} \\ &= -\left\langle \nabla g(\boldsymbol{x}^k), \nabla g(\boldsymbol{x}^k) \right\rangle \\ &= -\|\nabla g(\boldsymbol{x}^k)\|_2^2 = -\|\boldsymbol{A}^\mathsf{T}(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{b})\|_2^2 \end{aligned}$$

This implies that $g(\boldsymbol{x}^k - \tau \nabla g(\boldsymbol{x}^k)) \leq g(\boldsymbol{x}^k)$ when the step size τ is sufficiently small.

ii The sub-differential is given by

$$\partial h = \left\{ \begin{bmatrix} 1 \\ g_2 \end{bmatrix} : -4 \leq g_2 \leq 4 \right\}.$$

Hence

$$\boldsymbol{g}^{\max} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} &\lim_{\tau \to 0} \frac{h(\boldsymbol{x}^k - \tau \boldsymbol{g}^{\max}) - h(\boldsymbol{x}^k)}{\tau} \\ &= \lim_{\tau \to 0} \frac{(1 - \tau) + 4|0 - 4\tau| - (1 + 0)}{\tau} \\ &= 15 > 0. \end{aligned}$$

This implies that $h(\boldsymbol{x}^k - \tau \boldsymbol{g}^{\max}) > h(\boldsymbol{x}^k)$ when the step size τ is sufficiently small. Sub-gradient descent method may not converge in general.

(d) Let $\boldsymbol{x} \in \mathbb{R}^2$ and

$$f(x) = \underbrace{\frac{1}{2}\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2}_{g(\boldsymbol{x})} + |x_1| + 4|x_2|.$$

i At the point $\boldsymbol{x}^k$, find the first order Taylor approximation of $g(\boldsymbol{x})$, denoted by $\hat{g}(\boldsymbol{x})$. Write

$$\hat{g}(\boldsymbol{x}) + \frac{1}{2\tau}\|\boldsymbol{x} - \boldsymbol{x}^k\|_2^2$$

in a quadratic form of $\boldsymbol{x}$ (in terms of $\boldsymbol{A}$, $\boldsymbol{b}$, and τ). [2]

ii Find the optimal solution (in terms of $\boldsymbol{A}$, $\boldsymbol{b}$, and τ)

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} \ \hat{g}(\boldsymbol{x}) + \frac{1}{2\tau}\|\boldsymbol{x} - \boldsymbol{x}^k\|_2^2 + |x_1| + 4|x_2|.$$

[2]

Solution: The first order Taylor approximation of $g(\boldsymbol{x})$ is given by

$$\hat{g}(\boldsymbol{x}) = \frac{1}{2}\|\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{b}\|_2^2 + \left\langle \boldsymbol{A}^\mathsf{T}(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{b}), \boldsymbol{x} - \boldsymbol{x}^k \right\rangle.$$

Hence

$$\begin{aligned}
&\hat{g}(\boldsymbol{x}) + \frac{1}{2\tau}\|\boldsymbol{x} - \boldsymbol{x}^k\|_2^2 \\
&= \frac{1}{2\tau}\left\|\boldsymbol{x} - \boldsymbol{x}^k + \tau\boldsymbol{A}^\mathsf{T}(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{b})\right\|_2^2 + c \\
&= \frac{1}{2\tau}\left\|\boldsymbol{x} - \left(\boldsymbol{x}^k - \tau\boldsymbol{A}^\mathsf{T}(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{b})\right)\right\|_2^2 + c,
\end{aligned}$$

where c is a constant independent of $\boldsymbol{x}$. Therefore, the optimal solution is given by

$$\boldsymbol{x}^\star = \eta\left(\boldsymbol{x}^k - \tau\boldsymbol{A}^\mathsf{T}(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{b}); \begin{bmatrix}\tau \\ 4\tau\end{bmatrix}\right),$$

where η is the soft-thresholding function.

3. For a given vector $\boldsymbol{x} \in \mathbb{R}^{2n-1}$, an $n \times n$ Toeplitz matrix can be formed using the following mapping

$$\boldsymbol{Z} = \mathcal{T}(\boldsymbol{x}) := \begin{bmatrix} x_n & x_{n-1} & \cdots & x_1 \\ x_{n+1} & x_n & \cdots & x_2 \\ \vdots & \vdots & \ddots & \vdots \\ x_{2n-1} & x_{2n-2} & \cdots & x_n \end{bmatrix} \in \mathbb{R}^{n \times n}.$$

Or equivalently,

$$Z_{k,l} = x_{n+k-l}, \ 1 \leq k, l \leq n.$$

Given an index set $\Omega \subset \{1, 2, \cdots, 2n-1\}$, a truncated vector $\boldsymbol{y} = \boldsymbol{x}_\Omega$ is obtained by keeping all the elements in $\boldsymbol{x}$ indexed by Ω and deleting all elements indexed by $\Omega^c := \{1 \leq m \leq 2n-1 : m \notin \Omega\}$. Let $|\Omega|$ denote the size of the set Ω.

For any given matrix $\boldsymbol{A}$, let $\text{vec}(\boldsymbol{A})$ be the vector formed by stacking the columns of the matrix $\boldsymbol{A}$. Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{S}\boldsymbol{V}^\mathsf{T}$ denote its SVD.

(a) i Let $\|\cdot\|_*$ denote the nuclear norm. Write the closed form of the optimal solution

$$\boldsymbol{X}^\star = \arg\min_{\boldsymbol{X}} \ \|\boldsymbol{X}\|_* + \frac{\rho}{2}\|\boldsymbol{X} - \boldsymbol{B}\|_F^2,$$

where $\rho > 0$ is a constant and $\|\cdot\|_F^2$ is the squared Frobenius norm. No derivation is required here. [2]

ii Write $\boldsymbol{y} = \boldsymbol{x}_\Omega = \boldsymbol{P}\boldsymbol{x}$ where $\boldsymbol{P}$ is a matrix. Let $n = 3$ and $\Omega = \{1, 3, 4, 5\}$. Find the matrix $\boldsymbol{P}$ numerically. Compute the Moore-Penrose pseudo-inverse of $\boldsymbol{P}$ numerically, denoted by $\boldsymbol{P}^\dagger$. For a given vector $\boldsymbol{b}$ of proper length, find the optimal solution

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} \ \|\boldsymbol{P}\boldsymbol{x} - \boldsymbol{b}\|_2^2,$$

by using the numerical value of $\boldsymbol{P}$ that you found. [2]

A. For general n and Ω, and a given $\boldsymbol{b}$ of proper length, find the closed form solution for

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} \ \|\boldsymbol{P}\boldsymbol{x} - \boldsymbol{b}\|_2^2.$$

No derivation is required here. [2]

iii The linear transform $\boldsymbol{Z} = \mathcal{T}(\boldsymbol{x})$ can be written as $\boldsymbol{z} = \text{vec}(\boldsymbol{Z}) = \boldsymbol{T}\boldsymbol{x}$ for some matrix $\boldsymbol{T}$. Let $n = 3$ again. Find the matrix $\boldsymbol{T}$ numerically. Using the $\boldsymbol{T}$ that you found, compute the matrix $\boldsymbol{T}^\mathsf{T}\boldsymbol{T}$ numerically where the superscript T denotes matrix transpose. For a given vector $\boldsymbol{b}$ of proper length, find the closed form of the optimal solution

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} \ \|\boldsymbol{T}\boldsymbol{x} - \boldsymbol{b}\|_2^2,$$

by using the numerical value of $\boldsymbol{T}$ that you found. [3]

A. For general n and a given $\boldsymbol{B} \in \mathbb{R}^{n\times n}$, find the closed form solution for

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} \ \|\mathcal{T}(\boldsymbol{x}) - \boldsymbol{B}\|_F^2.$$

No derivation is required here. [4]

It is important to note that all the above three optimisation problems admit closed form solutions that only involving basic operations (no complicated matrix inverse).

Solution:

i

$$\boldsymbol{X}^\star = \boldsymbol{U}\text{Diagonal}\left(\left[\cdots, \eta(s_i; \frac{1}{\rho}), \cdots\right]^\mathsf{T}\right)\boldsymbol{V}^\mathsf{T},$$

where $\eta(s; 1/\rho) = \text{sign}(s)\max(0, |s| - 1/\rho)$ is the soft-thresholding function.

ii It is clear that

$$\boldsymbol{P} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \ \boldsymbol{P}^\dagger = \boldsymbol{P}^\mathsf{T} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

and

$$\boldsymbol{x}^\star = \begin{bmatrix} b_1, & 0, & b_2, & b_3, & b_4 \end{bmatrix}^\mathsf{T}.$$

A. In the general case,

$$\begin{cases} \boldsymbol{x}^\star_\Omega = \boldsymbol{b}, \\ \boldsymbol{x}^\star_{\Omega^c} = \boldsymbol{0}. \end{cases}$$

iii It can be verified that

$$\boldsymbol{T} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{9\times 5}.$$

Hence

$$\boldsymbol{T}^\mathsf{T}\boldsymbol{T} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and

$$\begin{aligned} \boldsymbol{x}^\star &= (\boldsymbol{T}^\mathsf{T}\boldsymbol{T})^{-1}\boldsymbol{T}^\mathsf{T}\boldsymbol{b} \\ &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \boldsymbol{T}^\mathsf{T}\boldsymbol{b}, \end{aligned}$$

which is given by

$$
\begin{aligned}
x_1^\star &= b_7 \\
x_2^\star &= (b_4 + b_8)/2 \\
x_3^\star &= (b_1 + b_5 + b_9)/3 \\
x_4^\star &= (b_2 + b_6)/2 \\
x_5^\star &= b_3.
\end{aligned}
$$

A. In general, it holds that

$$
\begin{aligned}
x_m^\star &= \begin{cases} \frac{1}{m} \sum_{\substack{m=n+k-l \\ 1\le k,l\le n}} B_{k,l} & \text{for } 1 \le m \le n, \\ \frac{1}{2n-m} \sum_{\substack{m=n+k-l \\ 1\le k,l\le n}} B_{l+(m-n),l} & \text{for } n < m \le 2n-1. \end{cases} \\
&= \begin{cases} \frac{1}{m} \sum_{k=1}^{m} B_{k,k+(n-m)} & \text{for } 1 \le m \le n, \\ \frac{1}{2n-m} \sum_{l=1}^{2n-m} B_{l+(m-n),l} & \text{for } n < m \le 2n-1. \end{cases}
\end{aligned}
$$

(b) Let us consider the following optimisation problem which arises in some applications:

$$
\begin{aligned}
&\min_{\boldsymbol{x}} \ \|\mathcal{T}(\boldsymbol{x})\|_* \\
&\text{s.t. } \boldsymbol{y} = \boldsymbol{x}_\Omega,
\end{aligned}
$$

where both $\boldsymbol{y}$ and Ω are given.

We will design an ADMM algorithm to solve the above constrained convex optimisation problem. Note that ADMM is designed to solve optimisation problem of the standard form

$$
\begin{aligned}
&\min_{\boldsymbol{x},\boldsymbol{z}} \ f(\boldsymbol{x}) + g(\boldsymbol{z}) \\
&\text{s.t. } \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c} = \boldsymbol{0}.
\end{aligned}
$$

i Design an ADMM formulation (in the standard form) such that sub-optimisation problems in the steps of ADMM admit closed-form solutions that you derived in Part (a). Specify the functions f and g. Specify $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{c}$ in terms of $\boldsymbol{P}$ and $\boldsymbol{T}$. Specify the dimensions of $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{c}$. [5]

ii Write the corresponding augmented Lagrangian where the penalty parameter is denoted by ρ and the Lagrange multiplier vector is denoted by $\boldsymbol{u}$. What is the dimension of $\boldsymbol{u}$? [2]

iii Write down the closed form solution to update the variable $\boldsymbol{x}$. Use the closed form solutions that you derived in Part (a) to get full marks. [2]

iv Write down the closed form solution to update the variable $\boldsymbol{z}$. Use the closed form solutions that you derived in Part (a) to get full marks. [2]

v Write down the closed form to update the Lagrange multiplier vector $\boldsymbol{u}$. [1]

Solution:

i A standard ADMM reformulation is given by

$$\begin{aligned}
&\min \ \underbrace{0}_{f} + \underbrace{\|\boldsymbol{Z}\|_*}_{g} \\
&\text{s.t.} \ \underbrace{\begin{bmatrix}\boldsymbol{T}\\ \boldsymbol{P}\end{bmatrix}}_{\boldsymbol{A}} \boldsymbol{x} + \underbrace{\begin{bmatrix}-\boldsymbol{I}\\ \boldsymbol{0}\end{bmatrix}}_{\boldsymbol{B}} \underbrace{\mathrm{vec}(\boldsymbol{Z})}_{\boldsymbol{z}} - \underbrace{\begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{y}\end{bmatrix}}_{\boldsymbol{c}} = \boldsymbol{0}.
\end{aligned}$$

Here $\boldsymbol{A} \in \mathbb{R}^{(n^2+|\Omega|)\times(2n-1)}$, $\boldsymbol{B} \in \mathbb{R}^{(n^2+|\Omega|)\times n^2}$, and $\boldsymbol{c} \in \mathbb{R}^{(n^2+|\Omega|)}$.

ii The augmetned Lagrangian is given by

$$\begin{aligned}
\mathcal{L} &= \|\boldsymbol{Z}\|_* + \langle \boldsymbol{u}, \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}\rangle + \frac{\rho}{2}\|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}\|_2^2 \\
&= \|\boldsymbol{Z}\|_* + \frac{\rho}{2}\|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c} + \frac{\boldsymbol{u}}{\rho}\|_2^2 + \text{Const.}
\end{aligned}$$

Here, $\boldsymbol{u} \in \mathbb{R}^{(n^2+|\Omega|)}$.

iii Let

$$\begin{aligned}
\boldsymbol{C}^z &= \boldsymbol{Z}^k - \text{reshape}(\frac{1}{\rho}\boldsymbol{c}^k_{1:n^2}, n, n) \\
\boldsymbol{c}^y &= \boldsymbol{y} - \frac{1}{\rho}\boldsymbol{c}^k_{n^2+1:n^2+|\Omega|}.
\end{aligned}$$

Then

$$x_m^{k+1} = \begin{cases} \frac{1}{m}\left(\sum_{k=1}^{m} C^z_{k,k+(n-m)}\right) & \text{for } 1 \le m \le n,\ m \notin \Omega, \\ \frac{1}{2n-m}\left(\sum_{l=1}^{2n-m} C^z_{l+(m-n),l}\right) & \text{for } n < m \le 2n-1,\ m \notin \Omega, \\ \frac{1}{m+1}\left(\sum_{k=1}^{m} C^z_{k,k+(n-m)} + c^y_{i_m}\right) & \text{for } 1 \le m \le n,\ m \in \Omega, \\ \frac{1}{2n-m+1}\left(\sum_{l=1}^{2n-m} C^z_{l+(m-n),l} + c^y_{i_m}\right) & \text{for } n < m \le 2n-1,\ m \in \Omega, \end{cases}$$

where i_m is defined such that $(\boldsymbol{P}\boldsymbol{x})_{i_m} = x_m$.

iv Let

$$\boldsymbol{C}^z = -\boldsymbol{T}\boldsymbol{x}^{k+1} - \text{reshape}(\frac{1}{\rho}\boldsymbol{c}^k_{1:n^2}, n, n).$$

Consider its SVD $\boldsymbol{C}^z = \boldsymbol{U}\boldsymbol{S}\boldsymbol{V}^\mathsf{T}$. Then

$$\boldsymbol{Z}^{k+1} = \boldsymbol{U}\text{Diagonal}\left(\left[\cdots, \eta(s_i; \frac{1}{\rho}), \cdots\right]^\mathsf{T}\right)\boldsymbol{V}^\mathsf{T}.$$

v

$$\boldsymbol{u}^{k+1} = \boldsymbol{u}^k + \rho(\boldsymbol{A}\boldsymbol{x}^{k+1} + \boldsymbol{B}\text{vec}(\boldsymbol{Z}^{k+1}) - \boldsymbol{c}).$$